

# AHMED ADHAM | RESUME

- » University: New Cairo Technological University
- » Skills: Python, C++/C#, Java, Data Structures & Algorithms, Arduino, Web Development
- » Interests: Quantum Computing, Machine Learning, Algorithms

## Summary

A passionate and driven Computer Science student with a strong interest in exploring groundbreaking technologies. Enthusiastic about Quantum Computing, Artificial Intelligence, and the Theory of Computation. Constantly seeking opportunities to learn, innovate, and apply knowledge to real-world challenges in the ever-evolving field of computer science.

## Education

- 2022 - Present
- Bachelor's Degree in Computer Science
- New Cairo Technological University
- » Major in Software Development.

## Experience

- Apr 2026
- Machine Learning Engineer, DeepX Hackathon
- Nile University, Egypt
- » Developed an Arabic Aspect-Based Sentiment Analysis (ABSA) system to analyze customer reviews across multiple aspects including food, service, and pricing.
- » Implemented multi-aspect sentiment classification pipelines and optimized preprocessing workflows for Arabic NLP tasks under a 12-hour hackathon deadline.
- » Introduced a *not\_mentioned* classification strategy to prevent forced sentiment predictions for absent aspects, significantly improving model accuracy and evaluation performance.
- » Worked on real-world Arabic customer review datasets using machine learning and NLP techniques for fine-grained sentiment understanding.
- Feb 2026
- Software Developer, Seshat(Healthcare Hackathon)
- Benha City, Egypt
- » Co-developed *Seshat*, a FHIR-compliant digital health platform transforming fragmented clinic systems into a connected ecosystem.
- » Implemented an AI-powered drug interaction engine, real-time lab results viewer, and *Wadjet* — an ICD-10 coding assistant to improve diagnostic accuracy and automate insurance workflows.
- » Focused on improving healthcare outcomes by reducing medication errors, accelerating diagnostics, and enabling continuity of care.
- Dec 2025
- Software Developer, GEM Hackathon (Hack054)
- Grand Egyptian Museum, Egypt
- » Developed a mobile application to support visitor experience analysis at the Grand Egyptian Museum using Bluetooth beacon-based indoor tracking.
- » Implemented a system to track visitor movement patterns, enabling data-driven insights into user behavior and museum engagement.
- » Designed an artifact bookmarking and recommendation feature to personalize visitor experiences based on user preferences.
- » Built a real-time backend system (*GEM Database Monitor*) to visualize live user interactions and continuously update analytics dashboards.
- Oct 2025
- Game Developer, Stellar Lands (NASA Space Apps Cairo 2025)
- Zewail City

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|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
|                     | <ul style="list-style-type: none"> <li>» Developed <i>Stellar Lands</i>, a farming simulator game built with the Godot engine for the NASA Space Apps Cairo 2025 challenge “<i>Farmer Navigator</i>”.</li> <li>» Implemented GIS-based terrain mapping and problem visualization to simulate environmental and agricultural conditions.</li> <li>» Designed mathematical growth models and simulation logic to realistically represent real-world planting dynamics.</li> <li>» Collaborated within a multidisciplinary team to integrate gameplay mechanics, balance systems, and deliver a polished, playable prototype by the submission deadline.</li> <li>» Created pixel-art visuals and interactive features to enhance scientific engagement and educational value.</li> </ul> |                                       |
| Sep 2025            | <b>Researcher, Quantum LSTM Air Quality Prediction</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Nile University                       |
|                     | <ul style="list-style-type: none"> <li>» Developed a Quantum LSTM (QLSTM) model to predict PM2.5 concentrations, validating results on real IBM quantum hardware with performance comparable to classical models.</li> <li>» Designed a hybrid GNN-LSTM spatio-temporal architecture to compress high-dimensional environmental data into efficient temporal representations.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                               |                                       |
| Jul 2025            | <b>AI Developer, ExamAi (EduHackAI-1, 3rd Place)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Sahaba Club                           |
|                     | <ul style="list-style-type: none"> <li>» Developed a bilingual smart exam monitoring system that enhances academic integrity through AI-based real-time proctoring.</li> <li>» Implemented facial and eye tracking using OpenCV, voice detection with SoundDevice, and activity logging with Pynput.</li> <li>» Integrated periodic snapshots, full-session recording, and real-time alerts with a multi-lingual (Arabic/English) GUI.</li> <li>» Built with Python, Tkinter, OpenCV, SoundDevice, Pandas, and Threading; tested successfully in a live exam simulation.</li> </ul>                                                                                                                                                                                                    |                                       |
| Dec 2024            | <b>Robotics Project</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | New Cairo Technological University    |
|                     | <ul style="list-style-type: none"> <li>» Built an Arduino-based line-following robot.</li> <li>» Handled both coding and physical construction, including wiring and sensor connections.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                       |
| Nov 2024            | <b>Project Management Trainee</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | American Chamber of Commerce in Egypt |
|                     | <ul style="list-style-type: none"> <li>» Completed a professional project management training focused on leadership, workflow optimization, and project lifecycle execution.</li> <li>» Gained hands-on experience in planning, coordination, and team management aligned with PMP best practices.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                       |
| Nov 2023 - Jul 2024 | <b>Frontend Developer, Instant Booking (2nd-Year College Project)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | New Cairo Technological University    |
|                     | <ul style="list-style-type: none"> <li>» Developed a responsive and user-friendly interface for a course booking platform.</li> <li>» Implemented animations and styling using CSS and Bootstrap.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                       |

## »»» Certifications

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|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| 2025 | <b>Machine Learning &amp; Python for AI(36 hrs)</b>                                                                                                                                                                                                                            | Black Horse Academy                    |
|      | <ul style="list-style-type: none"> <li>» Completed an applied course focused on both supervised and unsupervised machine learning techniques.</li> <li>» Worked on hands-on projects using Python, NumPy, Pandas, and TensorFlow for model training and evaluation.</li> </ul> |                                        |
| 2025 | <b>MEAN Stack Development (60 hrs)</b>                                                                                                                                                                                                                                         | Information Technology Institute (ITI) |

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|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
|                                          | <ul style="list-style-type: none"> <li>» Completed an intensive full-stack web development course covering MongoDB, Express.js, Angular, and Node.js.</li> <li>» Gained practical experience building dynamic web applications and RESTful APIs.</li> </ul>                                                                                                    |                                  |
| 2025                                     | <b>Quantum Summer School (2 Weeks)</b>                                                                                                                                                                                                                                                                                                                         | Nile University                  |
|                                          | <ul style="list-style-type: none"> <li>» Explored quantum computing concepts and their applications in machine learning.</li> <li>» Participated in a Quantum Machine Learning challenge focused on air quality prediction.</li> </ul>                                                                                                                         |                                  |
| 2024                                     | <b>Networking Principles</b>                                                                                                                                                                                                                                                                                                                                   | Radio Television Institute (RTI) |
|                                          | <ul style="list-style-type: none"> <li>» Studied computer networks, routing, switching, and troubleshooting techniques.</li> </ul>                                                                                                                                                                                                                             |                                  |
| 2024                                     | <b>Project Management Professional (PMP)</b>                                                                                                                                                                                                                                                                                                                   | American Chamber of Commerce     |
|                                          | <ul style="list-style-type: none"> <li>» Trained on leadership, project planning, workflow optimization, and professional project execution.</li> </ul>                                                                                                                                                                                                        |                                  |
| <b>»»» Activities &amp; Achievements</b> |                                                                                                                                                                                                                                                                                                                                                                |                                  |
| Apr 2026                                 | <b>DeepX Hackathon</b>                                                                                                                                                                                                                                                                                                                                         | Nile University, Egypt           |
|                                          | <ul style="list-style-type: none"> <li>» Achieved 11<sup>th</sup> place out of 151 competing teams in a 12-hour AI hackathon focused on Arabic Aspect-Based Sentiment Analysis (ABSA).</li> <li>» Built a fine-grained sentiment analysis system capable of identifying sentiments for specific review aspects rather than overall review polarity.</li> </ul> |                                  |
| Feb 2026                                 | <b>Benha Healthcare Hackathon</b>                                                                                                                                                                                                                                                                                                                              | Benha City, Egypt                |
|                                          | <ul style="list-style-type: none"> <li>» Co-founded <i>Seshat</i>, an AI-powered healthcare platform connecting clinics, doctors, and patients into a unified system.</li> <li>» Delivered a 20-minute pitch and achieved 2<sup>nd</sup> place in the medical track among competing teams.</li> </ul>                                                          |                                  |
| Dec 2025                                 | <b>GEM Hackathon (3rd Edition)</b>                                                                                                                                                                                                                                                                                                                             | Grand Egyptian Museum, Egypt     |
|                                          | <ul style="list-style-type: none"> <li>» Participated as part of team <i>Hack054</i>, developing a smart museum application for visitor tracking and personalized recommendations.</li> </ul>                                                                                                                                                                  |                                  |
| 2025                                     | <b>NASA Space Apps Cairo</b>                                                                                                                                                                                                                                                                                                                                   | Cairo, Egypt                     |
|                                          | <ul style="list-style-type: none"> <li>» Developed a playable prototype of <i>Stellar Lands</i> during the hackathon.</li> </ul>                                                                                                                                                                                                                               |                                  |
| 2025                                     | <b>EduHackAI-1</b>                                                                                                                                                                                                                                                                                                                                             | Dubai, UAE                       |
|                                          | <ul style="list-style-type: none"> <li>» Achieved 3<sup>rd</sup> place for the project <i>ExamAi</i>.</li> </ul>                                                                                                                                                                                                                                               |                                  |
| 2025                                     | <b>Quantum Summer School (2 Weeks)</b>                                                                                                                                                                                                                                                                                                                         | Nile University                  |
|                                          | <ul style="list-style-type: none"> <li>» Explored quantum computing concepts and their applications in machine learning.</li> <li>» Participated in a Quantum Machine Learning challenge focused on air quality prediction.</li> </ul>                                                                                                                         |                                  |
| 2024                                     | <b>Quantum AI Hackathon</b>                                                                                                                                                                                                                                                                                                                                    | Nile University                  |
|                                          | <ul style="list-style-type: none"> <li>» Worked on team-based quantum computing projects, developing innovative AI solutions.</li> </ul>                                                                                                                                                                                                                       |                                  |
| 2024                                     | <b>Google Developer Groups Cairo</b>                                                                                                                                                                                                                                                                                                                           | Cairo, Egypt                     |
|                                          | <ul style="list-style-type: none"> <li>» Attended workshops and hackathons, expanding practical knowledge in software development and AI.</li> </ul>                                                                                                                                                                                                           |                                  |